

**Exercises for the lecture
Relativity, Particles and Fields**

Summer semester 2026
Sheet 1

Lecture: Prof. Alejandro Ibarra.

Exercises:

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Group 1: Tue, Apr. 21, 8:30-10:00, Room PH 1121

Group 2: Tue, Apr 21, 12:00-14:00, Room PH 1121

Group 3: Wed, Apr 22, 12:00-14:00, Room PH 1121

Group 4: Thu, Apr 23, 10:00-12:00, C.3201

Problem 1 *Atmospheric muon lifetime*

Muons are unstable subatomic particles with an approximate mass of $m_\mu = 100 \text{ MeV}/c^2$ and a lifetime of $\tau = 1.5 \mu\text{s}$. They can be created when cosmic rays enter the Earth's atmosphere. Suppose that a muon is created at an altitude of 4 km above sea level and has an energy of 1000 MeV in the Earth rest frame. Its lifetime in its own rest frame is $1.5 \mu\text{s}$. Will the muon reach the sea level for

- [a] an observer in the Earth's rest frame?
- [b] an observer in the muon's rest frame?

Problem 2 *Length contraction and time dilation*

A train of proper length L enters a tunnel of proper length D , travelling at a velocity v .

- [a] Find the condition between L , D and v , such that the train is totally inside the tunnel at some time for an observer A standing outside the tunnel. What is the corresponding condition for an observer B inside the train?
- [b] When the rear end of the train enters the tunnel, observer B discovers that the tunnel exit is closed. B sends a laser signal from the rear end toward the exit in order to open it before the train arrives there. Show that the available reaction time τ_B in the rest frame of the observer B must satisfy

$$\tau_B < \frac{D\sqrt{1-\beta^2} - L(1+\beta)}{v}, \quad \beta = \frac{v}{c}.$$

with $\beta = v/c$.

Problem 3 *Ordering of spacetime events*

Three events A, B, C are seen by observer $\mathcal{O}$ to occur in the order ABC . Another observer, $\mathcal{O}'$, sees the events to occur in the order CBA . Can there always be a third observer who sees the events in the order ACB ? Restrict your considerations to one space dimension and support your conclusion by drawing a spacetime diagram.

Problem 4 *Acceleration in special relativity*

Let S' be an inertial frame which moves with velocity v in the x -direction with respect to S . In the frame S , the components of the acceleration of a particle are defined as

$$a_x = \frac{du_x}{dt}, \quad a_y = \frac{du_y}{dt}, \quad a_z = \frac{du_z}{dt}.$$

Express the components of the acceleration $a'_i = du'_i/dt'$ (with $i \in \{x, y, z\}$) in the frame S' in terms of u_i, a_i and v .

Part of the solution: $a'_x = \frac{(1 - v^2/c^2)^{3/2}}{(1 - u_x v/c^2)^3} a_x$

Problem 5 *Uniformly accelerating spaceship*

Consider a spaceship which is moving along the x -direction at a variable speed $u(t)$ relative to an inertial frame S . An observer inside the spaceship records an acceleration f in the x -direction, which is constant with respect to his own proper time τ .

[a] Using the result from problem 4, show that

$$\frac{du}{d\tau} = \left(1 - \frac{u^2}{c^2}\right) f$$

[b] Solve this differential equation, using the initial condition $u(\tau = 0) = 0$.

Hint: The solution is of the form $u(\tau) = c \tanh \xi(\tau)$ for some function $\xi(\tau)$.

[c] Derive the worldline of the spaceship with respect to the frame S , i.e. calculate $t(\tau)$ and $x(\tau)$. Discuss the solution using a spacetime diagram.